40nm Logic Salicide 0.9/1.0/1.1/1.2/1.8/2.5V Low leakage and Generic AA/Poly/Metal/Via layers Dummy Fill Script Release Note

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Version: v1.14_3

Version Data: 2017-06-01

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1.0 Scope

This document describes the Calibre rule-based dummy fill script of 40nm Logic Salicide 0.9/1.0/1.1/1.2/1.8/2.5V Low leakage and Generic AA/Poly/Metal/Via layers Calibre Rule Based Dummy Fill release. Switches selection will be discussed within.

1.1 Contents

Script file:

SMIC_CalDMF_40GELGLL_091011121825_V1.14_3.dmf

Release file:

SMIC_CalDMF_40GELGLL_091011121825_V1.14_3. pdf

1.2 Calibre rule-based dummy fill script option setup

a) This script fill AA, GT, M1~M8, V1-V7,TM1, TM2 , MTT2,ALPA layers. The switches in script are as below:

Note 1 : This script will select to fill dummy patterns for AA, GT, M1~M8, V1~V7, TM1, TM2, MTT2, ALPA layers according to user's setting .

Note 2 : Please select the filled dummy layers base on your layout.Set it to "YES", will fill the dummy pattern, otherwise set to "NO".

Note 3 : This script will insert two types metal dummy (MnDUM and MnDOP) for Mn, n = 1~8.

Note 4 : Please only set one switch to "Yes" between "TM2_insert" AND "MTT2_insert",while you must keep the other one to "NO".

Note 5 : Vn dummy patterns can be inserted only when the switches "Mn_insert", "Mn+1_insert" and "Vn_inserted" are set to "YES", n = 1~7.

Note 6 :"BORDER" layer must be drawn as "chip edge" in your layout in order to insert dummy patterns in it.

Note 7 : This runset must be run with hierarchy mode.

```
#DEFINE AA_insert YES
#DEFINE GT_insert YES
```

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```
#DEFINE M1_insert YES
#DEFINE V1_insert YES
#DEFINE M2_insert YES
#DEFINE V2_insert YES
#DEFINE M3_insert YES
#DEFINE V3_insert YES
#DEFINE M4_insert YES
#DEFINE V4_insert YES
#DEFINE M5_insert YES

#DEFINE V5_insert YES
#DEFINE M6_insert YES
#DEFINE V6_insert YES
#DEFINE M7_insert YES
#DEFINE V7_insert YES
#DEFINE M8_insert YES
#DEFINE TM1_insert YES

#DEFINE TM2_insert YES
#DEFINE MTT2_insert NO

#DEFINE ALPA_insert YES
```

a) Items should be taken care

A. It is the time to dummy fill, when your layout is finalized. Any changes of your layout after dummy fill, you should redo dummy fill under new layout without any old auto dummy fills remain.

B. It is forbidden to manually delete any shapes of auto dummy fills by script provided by SMIC. If you do not want to fill dummy at some area, you can draw dummy

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block layer to avoid it.

C. It is forbidden to copy any shapes of auto dummy fills by script provided by SMIC from one layout to another layout.

1.2 Tools and Tools Version

Calibre tool version: 2016.2_18.12

1.3 How to input commands to run utility

Recommend:

Calibre –64 –drc –hier –turbo 8

SMIC_CalDMF_40GELGLL_091011121825_V1.14_3.dmf | tee log.log

1.4 Calibre rule-based dummy fill solution output

After execute the command in 1.4, will get output file as below:

a) Output dummy file will only include added dummy layout with the name ---“*.gds”.You can change the gds name.

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